ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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PHYSICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Capacitance

English Medium

EXERCISE (O-1)

A. Capacitance of a system

1. Eight drops of mercury of equal radii possessing equal charges combine to form a big drop. Then the capacitance of bigger drop compared to each individual small drop is

(A) 8 times (B) 4 times

(C) 2 times (D) 32 times

CP0155

2. The distance between the plates of a parallel plate condenser is $4mm$ and potential difference is $60\ volts$. If the distance between the plates is increased to $12mm$, then

(A) The potential difference of the condenser will become $180\ volts$

(B) The P.D. will become $20\ volts$

(C) The P.D. will remain unchanged

(D) The charge on condenser will reduce to one third

CP0156

3. If 2μC charge is given to the one plate of a parallel plate capacitor of capacitance 1μF (initial charge on capacitor = 0). Potential difference across the plates of capacitor is

(A) 2V (B) 1V (C) $\frac{1}{2}$V (D) 4V

CP0157

4. A capacitor of capacitance C is charged to a potential V. The flux of the electric field through a closed surface enclosing the capacitor is

(A) $\frac{CV}{2\varepsilon_o}$ (B) $\frac{2CV}{\varepsilon_o}$ (C) $\frac{CV}{\varepsilon_o}$ (D) zero

CP0158

5. If n drops each of capacitance C coalesce to single drop, the capacitance of the single drop is

(A) C/n (B) nC (C) $n^{1/3}C$ (D) $n^{2/3}C$

CP0159

6. Two vertical metallic plates carrying equal and opposite charges are kept parallel to each other like a prallel plate capacitor. A small spherical metallic ball is suspended by a long insulated thread such that it hangs freely in thecentre of the two metallic plates. The ball, which is uncharged, is taken slowly towards the positively charged plate and is made to touch that plate. Then the ball will

(A) stick to the positively charged plate

(B) come back to its original position and will remain there

(C) oscillate between the two plates touching each plate in turn

(D) oscillate between the two plates without touching them

CP0160

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7. Choose the CORRECT statement :-

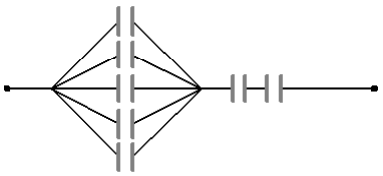
(A) C will increase on increasing Q (B) C will increase on decreasing Q

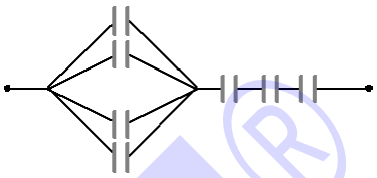
(C) C will increase on decreasing V (D) C doesnot depend on Q & V

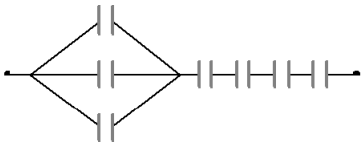
CP0031

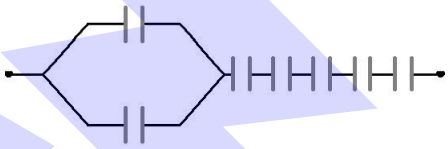
B. Capacitor circuits

8. Seven capacitors each of capacity $2\mu F$ are to be so connected to have a total capacity $\frac{10}{11}\mu F$. Which will be the necessary figure as shown

(A) 

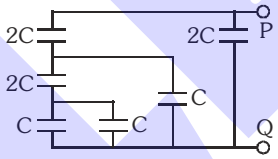
(B) 

(C) 

(D) 

CP0161

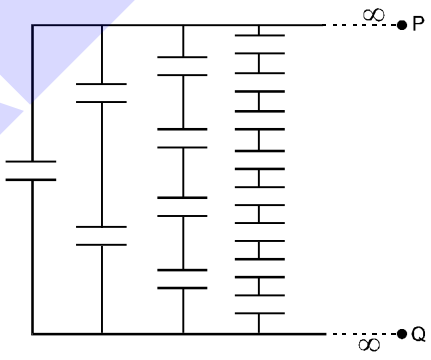
9. The value of equivalent capacitance of the combination shown in figure between the points P and Q is :



(A) 3 C (B) 2 C (C) C (D) C/3

CP0162

10. In the adjoining diagram if the capacity of each condenser is 1μF then the resultant capacity between the points P and Q will be -



(A) ∞ (B) zero (C) 2μF (D) 8μF

CP0163

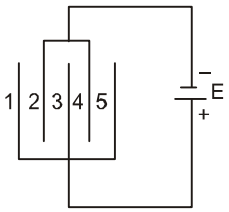
11. A parallel plate capacitor is made by stacking n similar metallic plates equally spaced from one another. The capacitance of the capacitor formed by any two neighbouring plates is C. The total capacitance of the combination will be

(A) $C/(n-1)$ (B) nC (C) $(n-1)C$ (D) $(n+1)C$

CP0164

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12. Five similar condenser plates, each of area A, are placed at equal distance d apart and are connected to a source of e.m.f. E as shown in the following diagram. The charge on the plates 1 and 4 will be-



(A) $\frac{\varepsilon_0 A}{d}, \frac{-2\varepsilon_0 A}{d}$

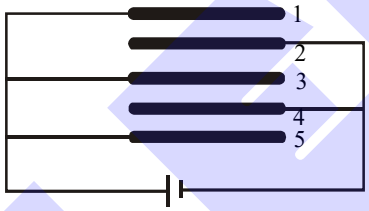
(B) $\frac{\varepsilon_0 AV}{d}, \frac{-2\varepsilon_0 AV}{d}$

(C) $\frac{-\varepsilon_0 AV}{d}, \frac{-3\varepsilon_0 AV}{d}$

(D) $\frac{\varepsilon_0 AV}{d}, \frac{-4\varepsilon_0 AV}{d}$

CP0165

13. Five identical plates are connected across a battery as follows



If the charge on plate 1 be +q, then the charges on the plates 2, 3, 4 & 5 are

(A) $-q, +q, -q, +q$

(B) $-2q, +2q, -2q, +q$

(C) $-q, +2q, -2q, +q$

(D) none of these

CP0166

14. From a supply of identical capacitors rated $8\,\mu F$ 250 V, the minimum number required to form a composite $16\mu F$ 1000V capacitor is

(A) 2 (B) 4 (C) 8 (D) 32

CP0167

15. A large sphere P of radius R is charged positively. It is momentarily connected to a small sphere Q of radius r. The two spheres now have same

(A) charge (B) electric field (C) energy (D) potential

CP0168

16. Two capacitors of capacitances C_1 and C_2 are connected in series across 200 V power supply. The potential drop across C_1 is 120V. A capacitor of capacitance $2\,\mu F$ is connected in parallel with C_1 and the potential drop across C_2 becomes 160 V. What are the values of C_1 and C_2 in μF ?

(A) 0.4 & 0.6 (B) 0.6 & 0.4 (C) 0.8 & 1.2 (D) 1.2 & 0.8

CP0169

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17. Two metallic charged spheres whose radii are $20\,cm$ and $10\,cm$ respectively, have each 150 $micro-coulomb$ positive charge. The common potential after they are connected by a conducting wire is

(A) 9×10^6 *volts* (B) 4.5×10^6 *volts* (C) 1.8×10^7 *volts* (D) 13.5×10^6 *volts*

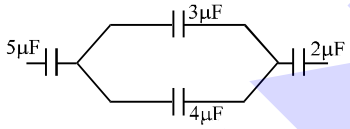
CP0170

18. A capacitor of capacitance $1\mu F$ can withstand a potential difference of 6 kV and another capacitor of $1\mu F$ can withstand a potential difference of 4V. If they are connected in series, the combination can withstand a potential difference of

(A) 3V (B) 4V (C) 6V (D) 8V

CP0171

19. If charge on left plate of the 5μF capacitor in the circuit segment shown in the figure is –20μC, the charge on the right plate of 3μF capacitor is :-



(A) +8.57 μC (B) –8.57 μC (C) +11.42 μC (D) –11.42 μC

CP0039

C. Energy analysis in capacitors

20. In a charged capacitor, the energy resides in

(A) The positive charges

(B) Both the positive and negative charges

(C) The field between the plates

(D) Around the edge of the capacitor plates

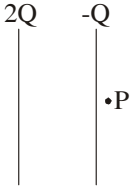
CP0172

21. A 40 μF capacitor in a defibrillator is charged to 3000 V. The energy stored in the capacitor is sent through the patient during a pulse of duration 2 ms. The power delivered to the patient is :–

(A) 45 kW (B) 90 kW (C) 180 kW (D) 360 kW

CP0173

22. In the figure shown the plates of a parallel plate capacitor have unequal charges. Its capacitance is 'C'. P is a point outside the capacitor and close to the plate of charge –Q. The distance between the plates is 'd' then which statement is wrong



(A) A point charge at point 'P' will experience electric force due to capacitor

(B) The potential difference between the plates will be $\frac{3Q}{2C}$

(C) The energy stored in the electric field in the region between the plates is $\frac{9Q^2}{8C}$

(D) The force on one plate due to the other plate is $\frac{Q^2}{2\pi\epsilon_0 d^2}$

CP0174

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23. A capacitor of capacity C is connected with a battery of potential V in parallel. The distance between its plates is reduced to half at once, assuming that the charge remains the same. Then to charge the capacitance upto the potential V again, the energy given by the battery will be

(A) $CV^2 / 4$ (B) $CV^2 / 2$ (C) $3CV^2 / 4$ (D) CV^2

CP0175

24. N identical capacitor are joined in parallel and the combination is charged to a potential V. Now if they are separated and then joined in series then energy of combination will :

(A) remain same and potential difference will also remain same

(B) remain same and potential difference will become NV

(C) increase N times and potential difference will become NV

(D) increase N time and potential difference will remains same

CP0176

25. A parallel plate capacitor is charged from a cell and then isolated from it. The separation between the plate is now increased

(A) the force of attraction between the plates will decrease

(B) the field in the region between the plates will change

(C) the energy stored in the capacitor will increase

(D) the potential difference between the plates will decreases

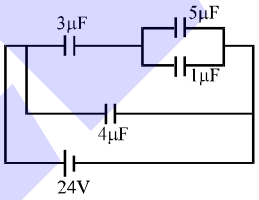
CP0177

26. A parallel-plate capacitor has a plate area of 0.3 m^2 and a plate separation of 0.1 mm. If the charge on each plate has a magnitude of 5×10^{-6} C then the force exerted by one plate on the other has a magnitude of about :

(A) 0 (B) 5 N (C) 1×10^4 N (D) 9×10^5 N

CP0052

27. In the circuit shown, the energy stored in 1μF capacitor is :-

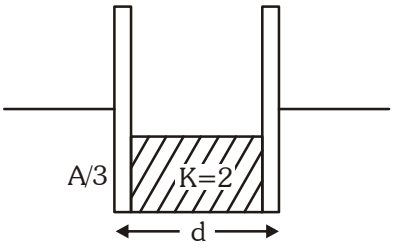


(A) 40 μJ (B) 64 μJ (C) 32 μJ (D) none

CP0053

D. Dielectrics

28. Capacitor of given system is C. Find the capacitance if dielectric is removed (A is the area of plate, K dielectric constant)

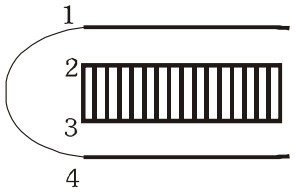


(A) $\frac{4}{3}C$ (B) $\frac{2}{3}C$ (C) $\frac{C}{3}$ (D) $\frac{3}{4}C$

CP0178

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29. Four identical plates 1,2,3 and 4 are placed parallel to each other at equal distance as shown in the figure. Plates 1 and 4 are joined together and the space between 2 and 3 is filled with a dielectric of dielectric constant k = 2. The capacitance of the system between 1 and 3 & 2 and 4 are C_1 and C_2 respectively. The ratio $\frac{C_1}{C_2}$ is-



(A) $\frac{5}{3}$ (B) 1 (C) $\frac{3}{5}$ (D) $\frac{5}{7}$

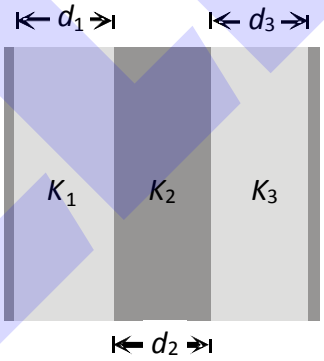
CP0179

30. A parallel plate capacitor has plates with area A and separation d. A battery charges the plates to a potential difference V_0. The battery is then disconnected and a dielectric slab of thickness d is introduced. The ratio of energy stored in the capacitor before and after slab is introduced, is:-

(A) K (B) $\frac{1}{K}$ (C) $\frac{A}{d^2K}$ (D) $\frac{d^2K}{A}$

CP0180

31. The expression for the capacity of the capacitor formed by compound dielectric placed between the plates of a parallel plate capacitor as shown in figure, will be (area of plate $= A$)



(A) $\frac{\varepsilon_0 A}{\left(\frac{d_1}{K_1}+\frac{d_2}{K_2}+\frac{d_3}{K_3}\right)}$ (B) $\frac{\varepsilon_0 A}{\left(\frac{d_1+d_2+d_3}{K_1+K_2+K_3}\right)}$ (C) $\frac{\varepsilon_0 A(K_1K_2K_3)}{d_1d_2d_3}$ (D) $\varepsilon_0\left(\frac{AK_1}{d_1}+\frac{AK_2}{d_2}+\frac{AK_3}{d_3}\right)$

CP0181

32. The capacity of a parallel plate condenser is C_0. If a dielectric of relative permitivity ε_r and thickness equal to one fourth the plate separation is placed between the plates, then its capacity becomes C. The value of $\frac{C}{C_0}$ will be:-

(A) $\frac{5\varepsilon_r}{4\varepsilon_r+1}$ (B) $\frac{4\varepsilon_r}{3\varepsilon_r+1}$ (C) $\frac{3\varepsilon_r}{2\varepsilon_r+1}$ (D) $\frac{2\varepsilon_r}{\varepsilon_r+1}$

CP0182

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33. The plates of a parallel plate capacitor are charged up to 100 volt. A 2 mm. thick dielectric plate is inserted between the plates, then to maintain the same potential difference, the distance between the capacitor plate is increased by 1.6 mm. The dielectric constant of the plate is :–

(A) 5 (B) 1.25 (C) 4 (D) 2.5

CP0183

34. A capacitor remains connected to a battery, a dielectric slab is slipped between the plates. The energy will increase due to

(A) increase in potential difference

(B) increase in electric field strength

(C) increase of capacitance

(D) none of above

CP0184

35. A parallel plate capacitor is connected across a 2V battery and charged. The battery is then disconnected and a glass slab is introduced between the plates. Which of the following pairs of quantities decrease?

(A) Charge and potential difference

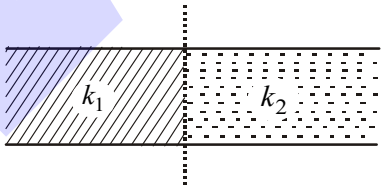
(B) Potential difference and energy stored

(C) Energy stored and capacitance

(D) Capacitance and charge

CP0185

36. A parallel plate capacitor with air as medium between the plates has a capacitance of $10\ \mu F$. The capacitor is divided into two halves and filled with two media as shown in the figure having dielectric constant $K_1 = 2$ *and* $K_2 = 4$. The capacitance of the system will now be



(A) $10\ \mu F$ (B) $20\ \mu F$ (C) $40\ \mu F$ (D) $30\ \mu F$

CP0186

37. A parallel plate capacitor having dielectric slab of $\varepsilon = 6$ is connected across a battery and charged. This dielectric slab is then removed and new dielectric slab of $\varepsilon_r = 10$ is introduced. The ratio of the energy stored in the first case to that in the second case is

(A) 3 : 5 (B) 5 : 3 (C) 9 : 25 (D) 25 : 9

CP0187

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38. The dielectric strength of a medium is $2kVmm^{-1}$. What is the maximum potential difference that can be set up across a $50\mu m$ specimen without puncturing it

(A) 10 V (B) 100 V
(C) 1000 V (D) 10,000 V

CP0188

39. A capacitor is connected to a battery. The electric energy stored in it is E. If the separation between the plates is doubled, what will be the energy on the capacitor ?

(A) 0.25 E (B) 0.50 E (C) E (D) 2 E

CP0189

40. A parallel plate capacitor has area of each plate as A, the separation between the plates as d and it is charged to potential V, and then disconnected from the battery. If a dielectric slab, completely filling the capacitor is introduced, how much work will be done in doing so

(A) $\frac{1}{2}\frac{V^2\varepsilon_0 A}{kd}$ (D) $\frac{1}{2}\frac{V^2\varepsilon_0 A}{k^2 d}$

(C) $\frac{1}{2}\frac{\varepsilon_0 AV^2}{d}\left(1-\frac{1}{k}\right)$ (D) $\frac{1}{2}\frac{\varepsilon_0 AV^2}{d}\left(1-\frac{1}{k^2}\right)$

CP0190

41. An air filled parallel plate capacitor charged to potential V_1 is connected to uncharged parallel plate capacitor having dielectric constant k. The common potential of both is V_2. What is the value of k ?

(A) $\frac{V_1-V_2}{V_1+V_2}$ (B) $\frac{V_1-V_2}{V_1}$ (C) $\frac{V_1-V_2}{V_2}$ (D) $\frac{V_1}{V_1-V_2}$

CP0191

42. An air-filled parallel-plate capacitor has a capacitance of 1 pF. The plate separation is then doubled and a wax dielectric is inserted, completely filling the space between the plates. As a result, the capacitance becomes 2 pF. The dielectric of the wax is

(A) 0.25 (B) 0.5 (C) 2.0 (D) 4.0

CP0059

43. A capacitor stores 60μC charge when connected across a battery. When the gap between the plates is filled with a dielectric, a charge of 120μC flows through the battery. The dielectric constant of the material inserted is :

(A) 1 (B) 2 (C) 3 (D) none

CP0192

44. Condenser A has a capacity of 15 μF when it is filled with a medium of dielectric constant 15. Another condenser B has a capacity 1 μF with air between the plates. Both are charged separately by a battery of 100V. After charging, both are connected in parallel without the battery and the dielectric material being removed. The common potential now is :-

(A) 400V (B) 800V (C) 1200V (D) 1600V

CP0063

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45. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of 30° with each other. When suspended in a liquid of density 0.8 g cm^{-3}, the angle remains the same. If density of the material of the sphere is 1.6 g cm^{-3}, the dielectric constant of the liquid is :

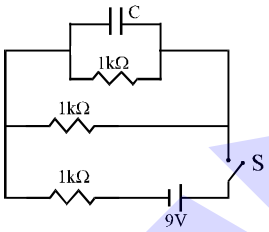
[AIEEE - 2010]

(A) 1 (B) 4 (C) 3 (D) 2

CP0101

E. R-C circuits

46. A capacitor C = 100 μF is connected to three resistor each of resistance 1 kΩ and a battery of emf 9V. The switch S has been closed for long time so as to charge the capacitor. When switch S is opened, the capacitor discharges with time constant :-



(A) 33 ms (B) 5 ms (C) 3.3 ms (D) 50 ms

CP0069

47. Let C be the capacitance of a capacitor discharging through a resistor R. Suppose t_1 is the time taken for the energy stored in the capacitor to reduce to half its initial value and t_2 is the time taken for the charge to reduce to one-fourth its initial value. Then the ratio t_1/t_2 will be : **[AIEEE-2010]**

(A) 2 (B) 1 (C) 1/2 (D) 1/4

CP0100

48. A resistor 'R' and 2μF capacitor in series is connected through a switch to 200 V direct supply. Across the capacitor is a neon bulb that lights up at 120 V. Calculate the value of R to make the bulb light up 5s after the switch has been closed. ($\log_{10} 2.5 = 0.4$) **[AIEEE-2011]**

(A) $2.7 \times 10^6\ \Omega$ (B) $3.3 \times 10^7\ \Omega$

(C) $1.3 \times 10^4\ \Omega$ (D) $1.7 \times 10^5\ \Omega$

CP0102

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EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

A. Capacitance of a system

1. A battery is used to charged a parallel-plate capacitor, after which it is disconnected. Then the plates are pulled apart to twice their original separation. This process will doubled the :

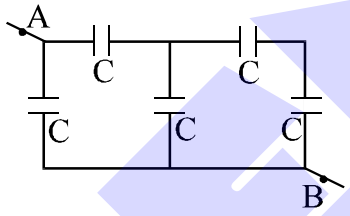
(A) capacitance (B) surface charge density on each plate

(C) stored energy (D) electric field between the two plates

CP0051

B. Capacitor circuits

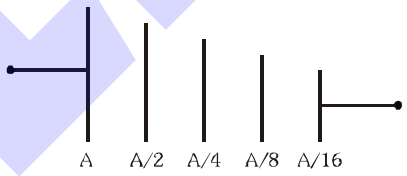
2. What is the equivalent capacitance of the system of capacitors between A & B as shown in the figure.



(A) $\frac{7}{6}C$ (B) 1.6 C (C) C (D) None

CP0040

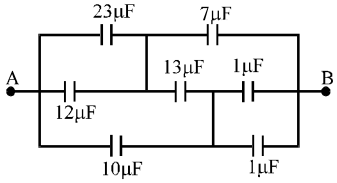
3. 5 Conducting plates each are placed face to face & equi-spaced at distance d. Area of each plate is half the previous plate. If area of first plate is A. Then the equivalent capacitance of the system shown is :-



(A) $\frac{\varepsilon_0 A}{d}$ (B) $\frac{\varepsilon_0 A}{10d}$ (C) $\frac{\varepsilon_0 A}{20d}$ (D) $\frac{\varepsilon_0 A}{30d}$

CP0041

4. Find the equivalent capacitance across A & B :-

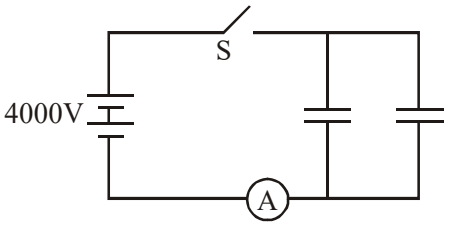


(A) $\frac{28}{3}$ μf (B) $\frac{15}{2}$ μF (C) 15 μF (D) none

CP0044

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\2-Electro-Grav-Current Elec.-Caps\Eng\04_Capacitance\01_[Eng].p65

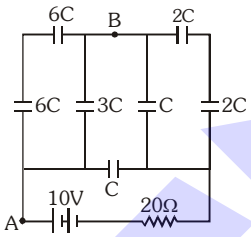
5. Each of the two 25-μF capacitors shown is initially unchanged. How many coulombs of charge pass through the ammeter A after the switch S is closed ?



(A) 0.10 (B) 0.20 (C) 10 (D) 0.05

CP0037

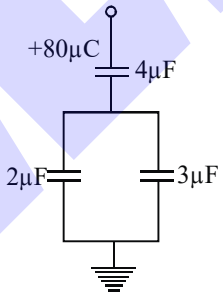
6. For the circuit shown here, the potential difference between points A and B is :-



(A) 2.5 V (B) 7.5 V (C) 10 V (D) Zero

CP0042

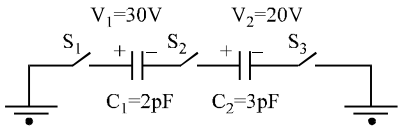
7. In the given circuit, a charge of +80 μC is given to the upper plate of the 4μF capacitor. Then in the steady state, the charge on the upper plate of the 3μF capacitor is :- **[IIT-JEE 2012]**



(A) + 32 μC (B) + 40μC (C) +48μC (D) + 80μC

CP0116

8. For the circuit shown, which of the following statements is true ? **[IIT-JEE 1999]**



(A) with S_1 closed, $V_1 = 15$ V, $V_2 = 20$ V
(B) with S_3 closed, $V_1 = V_2 = 25$ V
(C) with S_1 & S_2 closed, $V_1 = V_2 = 0$
(D) with S_1 & S_2 closed, $V_1 = 30$ V, $V_2 = 20$ V

CP0047

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Gravi-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

9. Three capacitors 2 μF, 3 μF and 5 μF can withstand voltages to 3V, 2V and 1V respectively. Their series combination can withstand a maximum voltage equal to :-

(A) 5 Volts (B) (31/6) Volts (C) (26/5) Volts (D) None

CP0064

C. Energy analysis in capacitors

10. Capacitors A and B are identical. Capacitor A is charged so it stores 4J of energy and capacitor B is uncharged. The capacitor are then connected in parallel. The total stored energy in the capacitors is now:

(A) 16 J (B) 8 J (C) 4 J (D) 2 J

CP0049

11. Consider a capacitor connected with a battery, capacitor is in steady state. Now plates of capacitor are drawn apart so as to double the separation in two cases.

Case :

(i) Battery remains connected

(ii) Battery is disconnected

Mark the **CORRECT** statement.

(A) In case (i) energy of capacitor increases

(B) In case (i) work done by battery is positive

(C) In case (ii) energy of capacitor increases

(D) In case (ii) potential difference across capacitor decreases

CP0055

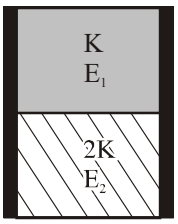
12. Two identical capacitors, have the same capacitance C. One of them is charged to potential V_1 and the other to V_2. The negative ends of the capacitors are connected together. When the positive ends are also connected, the decrease in energy of the combined system is :- **[IIT-JEE 2002 (Scr)]**

(A) $\frac{1}{4}C\left(V_1^2 - V_2^2\right)$ (B) $\frac{1}{4}C\left(V_1^2 + V_2^2\right)$ (C) $\frac{1}{4}C\left(V_1 - V_2\right)^2$ (D) $\frac{1}{4}C\left(V_1 + V_2\right)^2$

CP0056

D. Dielectrics

13. A parallel plate capacitor is connected from a cell and then isolated from it. Two dielectric slabs of dielectric constant K and 2K are now introduce in the region between upper half and lower half of the plate (as shown in figure). The electric field intensity in upper half of dielectric is E_1 and lower half is E_2 then



(A) $E_1 = 2E_2$

(B) Electrostatic potential energy of upper half is less than that of lower half

(C) Induced charges on both slabs are same

(D) Charge distribution on the plates remains same after insertion of dielectric

CP0065

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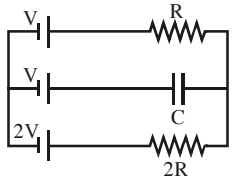
14. Two point charges exert a force F_0 on each other when placed in vacuum. Now the charges are increased to four times, separation between them is doubled and the system is placed is an insulating medium. Now they experience the same force. What should be the dielectric constant of the medium?

(A) 3 (B) 4 (C) 2 (D) 5

CP0066

E. R-C circuits

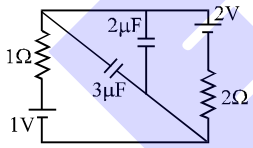
15. In the given circuit, with steady current the potential drop across the capacitor must be :-



(A) V (B) $\frac{V}{2}$ (C) $\frac{V}{3}$ (D) $\frac{2V}{3}$

CP0067

16. In the circuit shown, the charge on the 3μF capacitor at steady state will be



(A) 6 μC (B) 4 μC (C) $\frac{2}{3}$ μC (D) 3 μC

CP0068

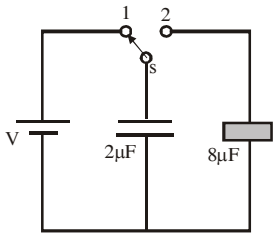
17. An uncharged capacitor of capacitance 4μF, a battery of emf 12 volt and a resistor of 2.5 MΩ are connected in series. The time after which $v_c = 3v_R$ is (take ln2 = 0.693) **[IIT-JEE' 2005 (Scr)]**

(A) 6.93 sec. (B) 13.86 sec.

(C) 20.52 sec. (D) none of these

CP0070

18. A 2 μF capacitor is charged as shown in figure. The percentage of its stored energy dissipated after the switch S is turned to position 2 is :- **[IIT-JEE 2011]**



(A) 0% (B) 20% (C) 75% (D) 80%

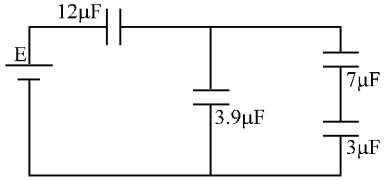
CP0115

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

MULTIPLE CORRECT TYPE QUESTIONS

B. Capacitor circuits

19. Four capacitors and a battery are connected as shown. The potential drop across the 7 μF capacitor is 6 V. Then the :



(A) potential difference across the 3 μF capacitor is 10 V

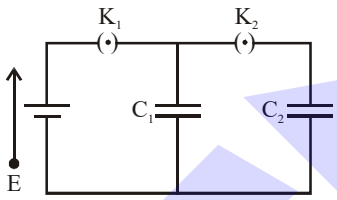
(B) charge on the 3 μF capacitor is 42 μC

(C) e.m.f. of the battery is 30 V

(D) potential difference across the 12 μF capacitor is 10 V.

CP0071

20. In the circuit shown in figure initially key K_1 is closed and key K_2 is open. Then K_1 is opened and K_2 is closed (order is important). [Take Q'_1 and Q'_2 as charges on C_1 and C_2 and V_1 and V_2 as voltage respectively. Then



(A) charge on C_1 gets redistributed such that $V_1 = V_2$

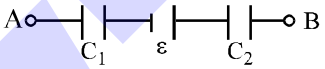
(B) charge on C_1 gets redistributed such that $Q'_1 = Q'_2$

(C) charge on C_1 gets redistributed such that $C_1V_1 + C_2V_2 = C_1E$

(D) charge on C_1 gets redistributed such that $Q'_1 + Q'_2 = Q$

CP0072

21. A circuit shown in the figure consists of a battery of emf 10 V and two capacitance C_1 and C_2 of capacitances 1.0 μF and 2.0 μF respectively. The potential difference $V_A - V_B$ is 5V



(A) charge on capacitor C_1 is equal to charge on capacitor C_2

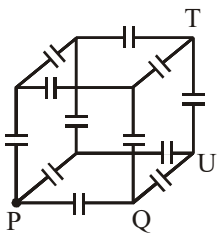
(B) Voltage across capacitor C_1 is 5V.

(C) Voltage across capacitor C_2 is 10 V

(D) Energy stored in capacitor C_1 is two times the energy stored in capacitor C_2.

CP0073

22. Twelve indentical capacitors each of capacitance C are connected as shown in figure.



(A) The effective capacitance between P and T is $\frac{6C}{5}$

(B) The effective capacitance between P and U is $\frac{4C}{3}$

(C) The effective capacitance between P and Q is $\frac{12C}{7}$

(D) All of the above statements are incorrect

CP0193

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav-Current Elec.-Caps\Eng\04_Capacitance\01_[Eng].p65

D. Dielectrics

23. A parallel plate air-core capacitor is connected across a source of constant potential difference. When a dielectric plate is introduced between the two plates then :
(A) some charge from the capacitor will flow back into the source.
(B) some extra charge from the source will flow back into the capacitor.
(C) the electric field intensity between the two plate does not change.
(D) the electric field intensity between the two plates will decrease.

CP0075

24. The capacitance of a parallel plate capacitor is C when the region between the plate has air. This region is now filled with a dielectric slab of dielectric constant k. The capacitor is connected to a cell of emf E, and the slab is taken out
(A) charge $CE(k-1)$ flows through the cell
(B) energy $E^2C(k-1)$ is absorbed by the cell.
(C) the energy stored in the capacitor is reduced by $E^2C(k-1)$
(D) the external agent has to do $\frac{1}{2}E^2C(k-1)$ amount of work to take the slab out.

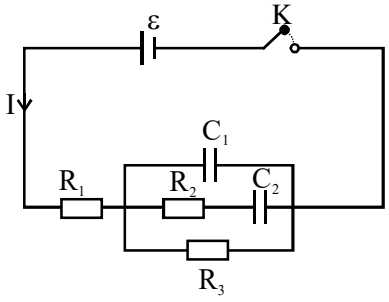
CP0077

25. A capacitor of capacity C_0 is connected to a battery of emf V_0. When steady state is attained a dielectric slab of dielectric constant K is slowly introduced in the capacitor. Mark the Correct statement(s), in final steady state :-
(A) Magnitude of induced charge on the each surface of slab is $C_0V_0(K-1)$
(B) Net electric force due to induced charges on the plate is zero.
(C) Force of attraction between plates of capacitor is $\frac{K(C_0V_0)^2}{2\epsilon_0 A}$
(D) Net field due to induced charges in dielectric slab is $\frac{8V_0(k-1)^2}{K\epsilon_0 A}$

CP0078

E. R-C circuits

26. Mark the **CORRECT** statement(s) regarding the current I through the battery in the circuit shown in figure.



(A) Immediately after the key K is closed, $I = \frac{\varepsilon}{R_1}$
(B) Immediately after the key K is closed, $I = \frac{\varepsilon}{R_1+R_3}$
(C) Long time after key K is closed, $I = \frac{\varepsilon}{R_1+R_3}$
(D) Long time after key K is closed, $I = \frac{\varepsilon}{R_1+R_2}$

CP0079

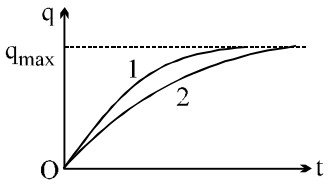
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COMPREHENSION TYPE QUESTIONS

E. R-C circuits

Paragraph for Question No. 27 & 28

The charge across the capacitor in two different RC circuits 1 and 2 are plotted as shown in figure.



27. Choose the correct statement(s) related to the two circuits.

(A) Both the capacitors are charged to the same charge.

(B) The emf's of cells in both the circuit are equal.

(C) The emf's of the cells may be different.

(D) The emf E_1 is more than E_2

CP0080

28. Identify the correct statement(s) related to the R_1, R_2, C_1 and C_2 of the two RC circuits.

(A) $R_1 > R_2$ if $E_1 = E_2$

(B) $C_1 < C_2$ if $E_1 = E_2$

(C) $R_1C_1 > R_2C_2$

(D) $\frac{R_1}{R_2} < \frac{C_2}{C_1}$

CP0081

MATRIX MATCH TYPE QUESTION

D. Dielectrics

29.

	Column-I		**Column-II**
(A)	Plates of an isolated, charged, parallel plate, air core capacitor are slowly pulled apart.	(P)	Electric energy stored inside capacitor increases in the process.
(B)	A dielectric is slowly inserted inside an isolated and charged parallel plate air cored capacitor to completely fill the space between plates.	(Q)	Force between the two plates of the capacitor remain unchanged.
(C)	Plates of a parallel plate capacitor connected across a battery are slowly pulled apart.	(R)	Electric field in the region between plates remain unchanged.
(D)	A dielectric slab is slowly inserted inside a parallel plate capacitor connected across a battery to completely fill the space between plates.	(S)	Total electric energy stored inside capacitor decreases in the process.
		(T)	Electric field in the region decreases.

CP0082

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\2 Electro-Grav+Current Elec.-Copa\Eng\04_Capacitance\01_[Eng].p65

EXERCISE (S)

A. Capacitance of a system

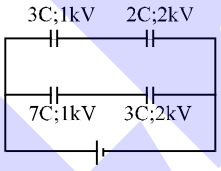
1. Two large parallel conducting plates are 34 mm apart and carry equal but opposite charges on their facing surfaces. An electron placed midway between the plates experiences a force of 3.2×10^{-16} N. What is the potential difference (in volts) between the plates?

CP0001

2. 2 conducting objects one with charge of +Q and another with –Q are kept on x-axis at x = 0 and x = 1 respectively. The electric field on the x-axis is given by $3Q\left(x^2+\frac{4}{3}\right)$. If the capacitance of this configuration of objects is C. Then fill $\frac{1}{C}$ (in F^{-1}) in OMR sheet.

CP0002

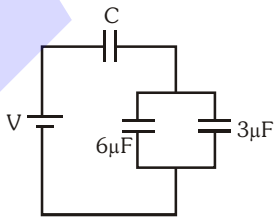
3. The diagram shows four capacitors with capacitances and break down voltages as mentioned. What should be the maximum value of the external emf source such that no capacitor breaks down?



CP0014

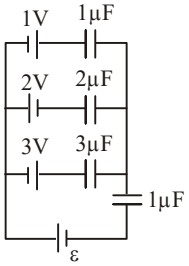
B. Capacitor circuits

4. If charge on 3μF capacitor is 3μC. Find the charge on capacitor of capacitance C in μC.



CP0004

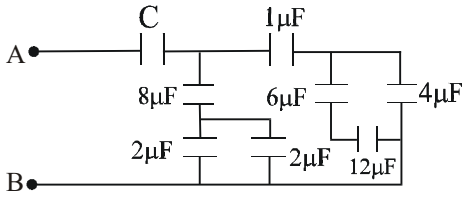
5. In the figure shown, find the e.m.f. ε for which charge on 2μF capacitor is 4 μC.



CP0005

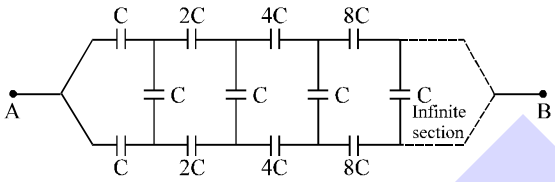
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Gravi-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

6. In the following circuit, the resultant capacitance between A and B is 1 μF. Find the value of C.



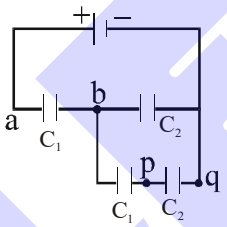
CP0006

7. Find the equivalent capacitance of the circuit between point A and B.



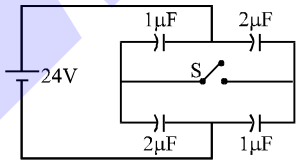
CP0007

8. In the given network if potential difference between p and q is 2V and $C_2 = 3C_1$. Then find the potential difference between a & b.



CP0008

9. The connections shown in figure are established with the switch S open. How much charge will flow through the switch if it is closed ?

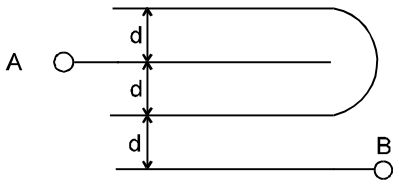


CP0010

10. The plates of a parallel plate capacitor are given charges +4Q and –2Q. The capacitor is then connected across an uncharged capacitor of same capacitance as first one (= C). Find the final potential difference between the plates of the first capacitor.

CP0011

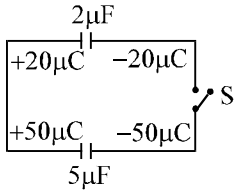
11. Find out equivalent capacitance between A and B. (Given area of plate is A)



node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2.Electro-Grav-Current Elec.-Caps\Eng\04_Capacitance\01_[Eng].p65

C. Energy analysis in capacitors

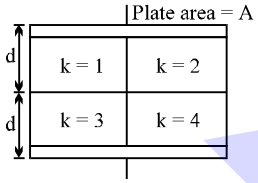
12. Find heat produced in the circuit shown in figure on closing the switch S.



CP0009

D. Dielectrics

13. Find the capacitance of the system shown in figure.

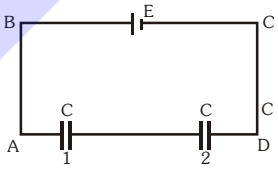


CP0012

14. Two square metallic plates of 1 m side are kept 0.01 m apart, like a parallel plate capacitor, in air in such a way that one of their edges is perpendicular, to an oil surface in a tank filled with an insulating oil. The plates are connected to a battery of e.m.f. 500 volt . The plates are then lowered vertically into the oil at a speed of 0.001 m/s. Calculate the current drawn from the battery during the process. [di-electric constant of oil = 11, $\in_0 = 8.85 \times 10^{-12}$ $C^2/N^2 m^2$]

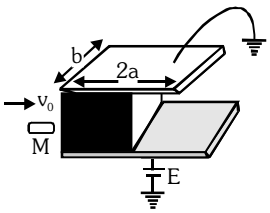
CP0194

15. In the adjoining figure, capacitor (1) and (2) have a capacitance 'C' each. When the dielectric of dielectric constant K is inserted between the plates of one of the capacitor, find the total charge flowing through battery.



CP0195

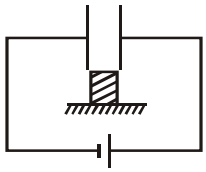
16. A parallel plate capacitor is half filled with a dielectric (K) of mass M. Capacitor is attached with a cell of emf E. Plates are held fixed on smooth insulating horizontal surface. A bullet of mass M hits the dielectric elastically and its found that dielectric just leaves out the capacitor. Find speed of bullet.



CP0196

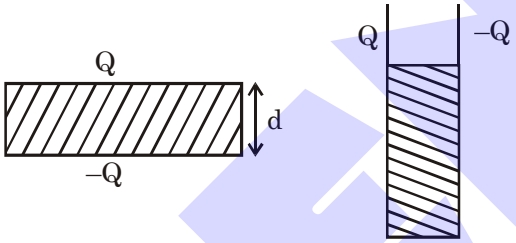
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Gravi-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

17. A dielectric slab of constant k = 2, area 10cm × 10cm and width 0.885 mm is just kept below a capacitor having same dimensions. The slab has mass of 0.5 mg. If the normal force exerted by the surface is zero, find the voltage of the battery.



CP0197

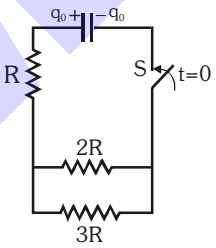
18. A isolated capacitor of charge Q and plate area L × L kept at a distance d from each other is held horizontally. It is completely filled with dielectric of density ρ and dielectric constant k. When it is made vertical, the dielectric starts moving out and comes to rest at half the width. Find Q in terms of other parameters.(k = 2)



CP0198

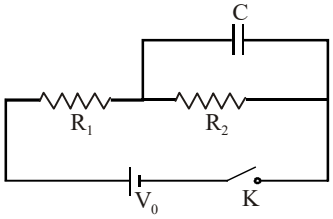
E. R-C circuits

19. At t = 0 charge on capacitor is q_0. Now switch S is closed. Heat loss in 3R is x × 10^{-6} J. Then find the value of x. [Given q_0= 15μC, C = 6/55 μF]



CP0016

20. In the connection shown in the *figure* the switch K is open and the capacitor is uncharged. Then we close the switch and let the capacitor charge up to the maximum and open the switch again. Then (Use the following data : V_0 = 30 V, R_1 = 10 kΩ, R_2 = 5 kΩ.)



(i) the current through R_1 be I_1 immediately after closing the switch

(ii) the current through R_2 be I_2 a long time after the switch was closed

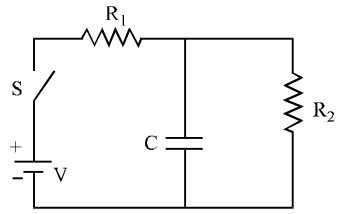
(iii) the current through R_2 be I_3 immediately after reopening the switch

Find the value of $\frac{I_1}{I_2 I_3}$ (in ampere^{-1}).

CP0017

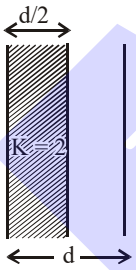
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\2-Electro-Grav-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

21. In the given circuit, the switch S is closed at time t = 0. The charge Q on the capacitor at any instant t is given by $Q(t) = Q_0 (1 - e^{-\alpha t})$. Find the value of Q_0 and α in terms of given parameters shown in the circuit. **[IIT-JEE 2005]**



CP0018

22. A certain series RC circuit is formed using a resistance R, a capacitor without dielectric having a capacitance C = 2F and a battery of emf E = 3V. The circuit is completed and it is allowed to attain the steady state. After this, at t = 0, half the thickness of the capacitor is filled with a dielectric of constant K = 2 as shown in the figure. The system is again allowed to attain a steady state. What will be the heat generated (in joule) in the circuit between t = 0 and t = ∞ ?

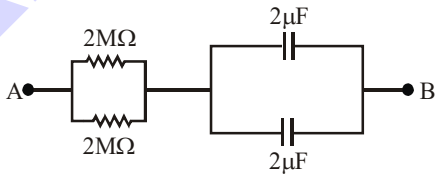


CP0019

23. A capacitor filled with dielectric of permittivity ε = 2.1 loses half the charge acquired during a time interval τ = 3.0 min. Assuming the charge to leak only through the dielectric filler, calculate its resistivity.

CP0020

24. At time t = 0, a battery of 10V is connected across points A and B in the given circuit. If the capacitors have no charge initially, at what time (in seconds) does the voltage across them become 4V?
[Take : $\ell n\, 5 = 1.6$, $\ell n 3 = 1.1$] **[IIT-JEE 2010]**



CP0114

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Gravi-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

EXERCISE (J-M)

1. Two capacitors C_1 and C_2 are charged to 120V and 200V respectively. It is found that by connecting them together the potential on each one can be made zero. Then : **[JEE-Main-2013]**

(1) $5C_1 = 3C_2$ (2) $3C_1 = 5C_2$ (3) $3C_1 + 5C_2 = 0$ (4) $9C_1 = 4C_2$

CP0105

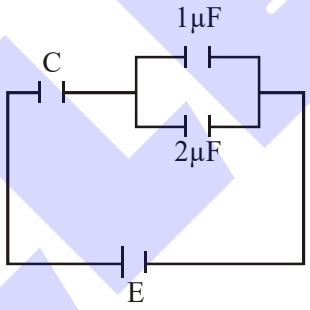
2. A parallel plate capacitor is made of two circular plates separated by a distance of 5 mm and with a dielectriic of dielectric constant 2.2 between them. When the electric field in the dielectric is 3×10^4 V/m, the charge density of the positive plate will be close to : **[JEE-Main-2014]**

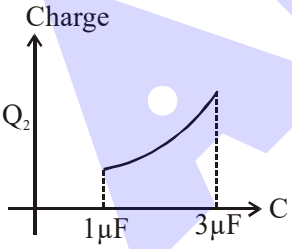
(1) 3×10^4 C/m^2 (2) 6×10^4 C/m^2 (3) 6×10^{-7} C/m^2 (4) 3×10^{-7} C/m^2

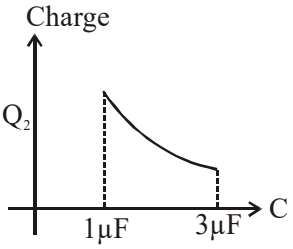
CP0106

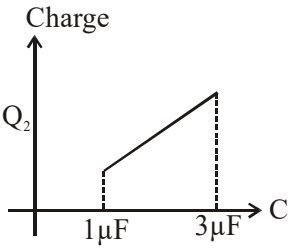
3. In the given circuit, charge Q_2 on the 2μF capacitor changes as C is varied from 1μF to 3μF. Q_2 as a function of 'C' is given properly by : (figures are drawn schematically and are not to scale) :-

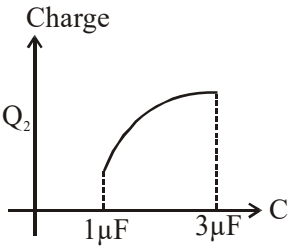
[JEE-Main-2015]



(1)


(2)


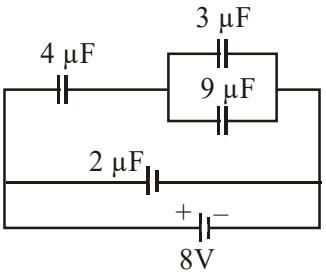
(3)


(4)


CP0107

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

4. A combination of capacitors is set up as shown in the figure. The magnitude of the electric field, due to a point charge Q (having a charge equal to the sum of the charges on the 4 μF and 9 μF capacitors), at a point 30 m from it, would equal: **[JEE-Main-2016]**



(1) 480 N/C (2) 240 N/C (3) 360 N/C (4) 420 N/C

CP0108

5. Three capacitors each of 4 μF are to be connected in such a way that the effective capacitance is 6 μF. This can be done by connecting them : **[JEE-Main online-2016]**

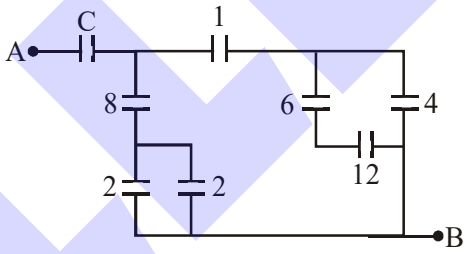
(1) two in parallel and one in series (2) all in parallel

(3) two in series and one in parallel (4) all in series

CP0109

6. Figure shows a network of capacitors where the numbers indicates capacitances in micro Farad. The value of capacitance C if the equivalent capacitance between point A and B is to be 1 μF is :-

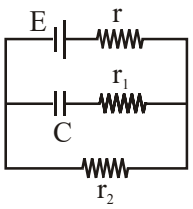
[JEE-Main online-2016]



(1) $\frac{33}{23}\mu F$ (2) $\frac{34}{23}\mu F$ (3) $\frac{31}{23}\mu F$ (4) $\frac{32}{23}\mu F$

CP0110

7. In the given circuit diagram when the current reaches steady state in the circuit, the charge on the capacitor of capacitance C will be : **[JEE-Main-2017]**



(1) $CE\frac{r_2}{(r+r_2)}$ (2) $CE\frac{r_1}{(r_1+r)}$ (3) CE (4) $CE\frac{r_1}{(r_2+r)}$

CP0111

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Gravi-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

8. A capacitance of 2 μF is required in an electrical circuit across a potential difference of 1.0 kV. A large number of 1 μF capacitors are available which can withstand a potential difference of not more than 300 V. The minimum number of capacitors required to achieve this is : **[JEE-Main-2017]**

(1) 24 (2) 32 (3) 2 (4) 16

CP0112

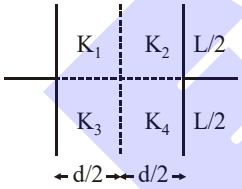
9. A parallel plate capacitor of capacitance 90 pF is connected to a battery of emf 20V. If a dielectric material of dielectric constant $K = \frac{5}{3}$ is inserted between the plates, the magnitude of the induced charge will be :- **[JEE-Main-2018]**

(1) 0.3 n C (2) 2.4 n C (3) 0.9 n C (4) 1.2 n C

CP0113

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

10. A parallel plate capacitor with square plates is filled with four dielectrics of dielectric constants K_1, K_2, K_3, K_4 arranged as shown in the figure. The effective dielectric constant K will be : **[JEE-Main-2019_Jan]**

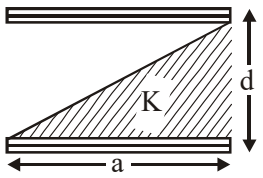


(1) $K = \frac{(K_1 + K_2)(K_3 + K_4)}{2(K_1 + K_2 + K_3 + K_4)}$ (2) $K = \frac{(K_1 + K_2)(K_3 + K_4)}{(K_1 + K_2 + K_3 + K_4)}$

(3) $K = \frac{(K_1 + K_4)(K_2 + K_3)}{2(K_1 + K_2 + K_3 + K_4)}$ (4) $K = \frac{(K_1 + K_3)(K_2 + K_4)}{K_1 + K_2 + K_3 + K_4}$

CP0134

11. A parallel plate capacitor is made of two square plates of side 'a', separated by a distance d (d<<a). The lower triangular portion is filled with a dielectric of dielectric constant K, as shown in the figure. Capacitance of this capacitor is : **[JEE-Main-2019_Jan]**



(1) $\frac{1}{2}\frac{k \in_0 a^2}{d}$ (2) $\frac{k \in_0 a^2}{d} \ln K$ (3) $\frac{k \in_0 a^2}{d(K-1)} \ln K$ (4) $\frac{k \in_0 a^2}{2d(K+1)}$

CP0135

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

12. A parallel plate capacitor has 1μF capacitance. One of its two plates is given +2μC charge and the other plate, +4μC charge. The potential difference developed across the capacitor is:-

[JEE-Main-2019_April]

(1) 5V (2) 2V

(3) 3V (4) 1V

CP0136

13. The parallel combination of two air filled parallel plate capacitors of capacitance C and nC is connected to a battery of voltage, V. When the capacitors are fully charged, the battery is removed and after that a dielectric material of dielectric constant K is placed between the two plates of the first capacitor. The new potential difference of the combined system is :- **[JEE-Main-2019_April]**

(1) $\frac{V}{K+n}$ (2) V (3) $\frac{(n+1)V}{(K+n)}$ (4) $\frac{nV}{K+n}$

CP0137

14. A capacitor with capacitance 5μF is charged to 5μC. If the plates are pulled apart to reduce the capacitance to 2μF, how much work is done ? **[JEE-Main-2019_April]**

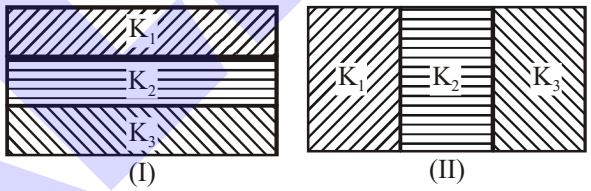
(1) 3.75×10^{-6} J (2) 2.55×10^{-6} J

(3) 2.16×10^{-6} J (4) 6.25×10^{-6} J

CP0138

15. Two identical parallel plate capacitors, of capacitance C each, have plates of area A, separated by a distance d. The space between the plates of the two capacitors, is filled with three dielectrics, of equal thickness and dielectric constants K_1, K_2 and K_3. The first capacitor is filled as shown in fig. I, and the second one is filled as shown in fig. II.

If these two modified capacitors are charged by the same potential V, the ratio of the energy stored in the two, would be (E_1 refers to capacitor (I) and E_2 to capacitor (II)) : **[JEE-Main-2019_April]**



(1) $\frac{E_1}{E_2} = \frac{9K_1K_2K_3}{(K_1+K_2+K_3)(K_2K_3+K_3K_1+K_1K_2)}$

(2) $\frac{E_1}{E_2} = \frac{K_1K_2K_3}{(K_1+K_2+K_3)(K_2K_3+K_3K_1+K_1K_2)}$

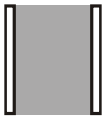
(3) $\frac{E_1}{E_2} = \frac{(K_1+K_2+K_3)(K_2K_3+K_3K_1+K_1K_2)}{K_1K_2K_3}$

(4) $\frac{E_1}{E_2} = \frac{(K_1+K_2+K_3)(K_2K_3+K_3K_1+K_1K_2)}{9K_1K_2K_3}$

CP0139

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Gravi-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

16. A parallel plate capacitor has plates of area A separated by distance 'd' between them. It is filled with a dielectric which has a dielectric constant that varies as $k(x) = K(1 + \alpha x)$ where 'x' is the distance measured from one of the plates. If $(\alpha d) << 1$, the total capacitance of the system is best given by the expression : **[JEE-Main-2020_Jan]**



(1) $\frac{AK\varepsilon_0}{d}\left(1+\frac{\alpha d}{2}\right)$

(2) $\frac{A\varepsilon_0 K}{d}\left(1+\left(\frac{\alpha d}{2}\right)^2\right)$

(3) $\frac{A\varepsilon_0 K}{d}\left(1+\frac{\alpha^2 d^2}{2}\right)$

(4) $\frac{AK\varepsilon_0}{d}(1+\alpha d)$

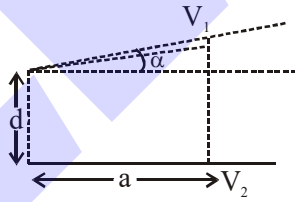
CP0140

17. Effective capacitance of parallel combination of two capacitors C_1 and C_2 is 10 μF. When these capacitors are individually connected to a voltage source of 1V, the energy stored in the capacitor C_2 is 4 times that of C_1. If these capacitors are connected in series, their effective capacitance will be : **[JEE-Main-2020_Jan]**

(1) 3.2 μF (2) 8.4 μF (3) 1.6 μF (4) 4.2 μF

CP0141

18. A capacitor is made of two square plates each of side 'a' making a very small angle α between them, as shown in figure. The capacitance will be close to : **[JEE-Main-2020_Jan]**



(1) $\frac{\varepsilon_0 a^2}{d}\left(1-\frac{3\alpha a}{2d}\right)$

(2) $\frac{\varepsilon_0 a^2}{d}\left(1-\frac{\alpha a}{4d}\right)$

(3) $\frac{\varepsilon_0 a^2}{d}\left(1+\frac{\alpha a}{d}\right)$

(4) $\frac{\varepsilon_0 a^2}{d}\left(1-\frac{\alpha a}{2d}\right)$

CP0142

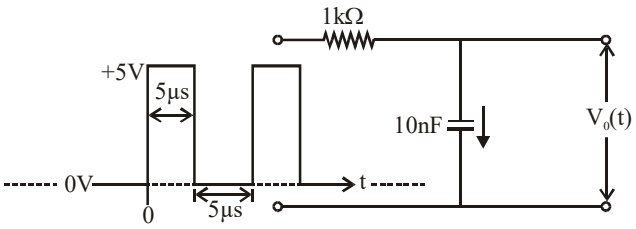
19. A parallel plate capacitor has plate of length '*l*', width 'w' and separation of plates is 'd'. It is connected to a battery of emf V. A dielectric slab of the same thickness 'd' and of dielectric constant k = 4 is being inserted between the plates of the capacitor. At what length of the slab inside plates, will be energy stored in the capacitor be two times the initial energy stored? **[JEE-Main-2020_Sep]**

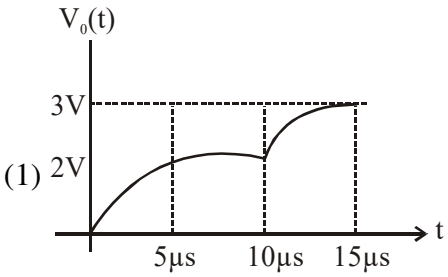
(1) $l/4$ (2) $l/2$ (3) $l/3$ (4) $2l/3$

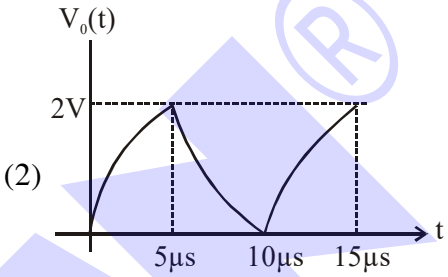
CP0143

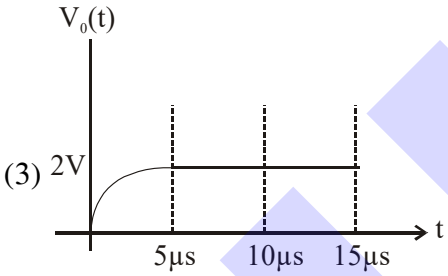
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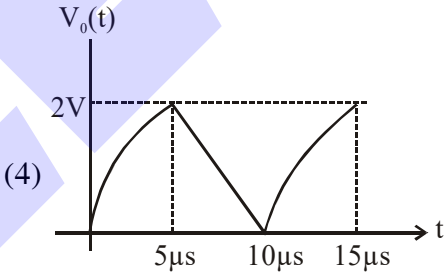
20. For the given input voltage waveform $V_{in}(t)$, the output voltage waveform $V_D(t)$, across the capacitor is correctly depicted by: **[JEE-Main-2020_Sep]**



(1)


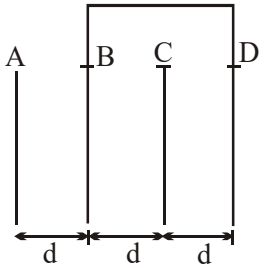
(2)


(3)


(4)


CP0144

21. Four identical rectangular plates with length, $l = 2$ cm and breadth, $b = \frac{3}{2}$ cm are arranged as shown in figure. The equivalent capacitance between A and C is $\frac{x\varepsilon_0}{d}$. The value of x is ___. (Round off to the Nearest Integer) **[JEE-Main-2021_March]**



CP0199

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Gravi-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

22. A parallel plate capacitor with plate area 'A' and distance of separation 'd' is filled with a dielectric. What is the capacity of the capacitor when permittivity of the dielectric varies as :

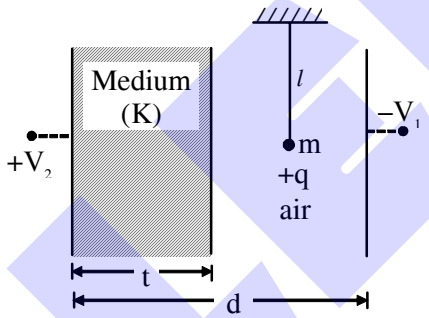
$\varepsilon(x) = \varepsilon_0 + kx$, for $\left(0 < x \le \frac{d}{2}\right)$ **[JEE-Main-2021_July]**

$\varepsilon(x) = \varepsilon_0 + k(d-x)$, for $\left(\frac{d}{2} \le x \le d\right)$

(1) $\left(\varepsilon_0 + \frac{kd}{2}\right)^{2/kA}$ (2) $\frac{kA}{2\ln\left(\frac{2\varepsilon_0 + kd}{2\varepsilon_0}\right)}$ (3) 0 (4) $\frac{kA}{2}\ln\left(\frac{2\varepsilon_0}{2\varepsilon_0 - kd}\right)$

CP0200

23. A simple pendulum of mass 'm', length '*l*' and charge '+q' suspended in the electric field produced by two conducting parallel plates as shown. The value of deflection of pendulum in equilibrium position will be **[JEE-Main-2021_July]**

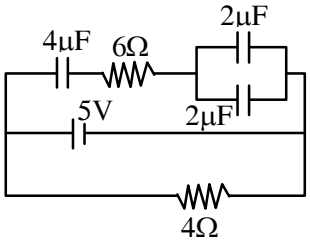


(1) $\tan^{-1}\left[\frac{q}{mg} \times \frac{C_1(V_2 - V_1)}{(C_1 + C_2)(d-t)}\right]$ (2) $\tan^{-1}\left[\frac{q}{mg} \times \frac{C_2(V_2 - V_1)}{(C_1 + C_2)(d-t)}\right]$

(3) $\tan^{-1}\left[\frac{q}{mg} \times \frac{C_2(V_1 + V_2)}{(C_1 + C_2)(d-t)}\right]$ (4) $\tan^{-1}\left[\frac{q}{mg} \times \frac{C_1(V_1 + V_2)}{(C_1 + C_2)(d-t)}\right]$

CP0201

24. Calculate the amount of charge on capacitor of 4 μF. The internal resistance of battery is 1Ω : **[JEE-Main-2021_Aug]**



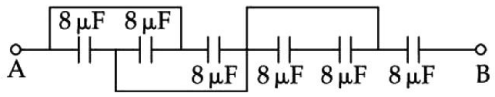
(1) 8 μC (2) zero (3) 16 μC (4) 4 μC

CP0202

node06\B09C-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Grav-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

25. The equivalent capacitance between points A and B in below shown figure will be _____μF.

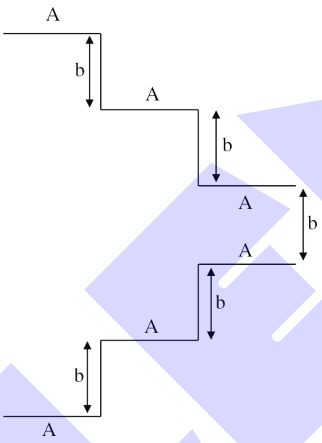
[JEE-Main-2022_June]



CP0203

26. A parallel plate capacitor is made up of stair like structure with a palte area A of each stair and that is connected with a wire of length b, as shown in the figure. The capacitance of the arrangement is $\frac{x}{15}\frac{\varepsilon_0 A}{b}$. The value of x is _____ .

[JEE-Main-2022_June]



CP0204

27. Capacitance of an isolated conducting sphere of radius R_1 becomes n times when it is enclosed by a concentric conducting sphere of radius R_2 connected to earth. The ratio of their radii $\left(\frac{R_2}{R_1}\right)$ is:

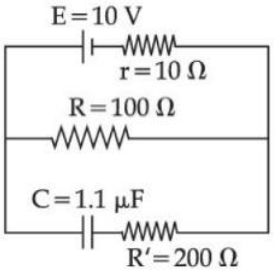
[JEE-Main-2022_July]

(A) $\frac{n}{n-1}$ (B) $\frac{2n}{2n+1}$ (C) $\frac{n+1}{n}$ (D) $\frac{2n+1}{n}$

CP0205

28. As show in the figure, in steady state, the charge stored in the capacitor is……. $\times 10^{-6}$C.

[JEE-Main-2022_July]

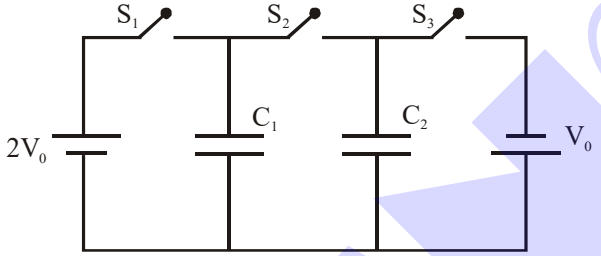


CP0206

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Gravi-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

EXERCISE (J-A)

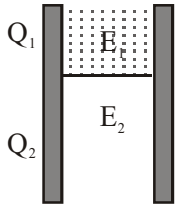
1. In the circuit shown in the figure, there are two parallel plate capacitors each of the capacitance C. The switch S_1 is pressed first to fully charge the capacitor C_1 and then released. The switch S_2 is then pressed to charge the capacitor C_2. After some time, S_2 is released and then S_3 is pressed, After some time, **[JEE-Advance 2013]**



(A) the charge on the upper plate of C_1 is $2CV_0$ (B) the charge on the upper plate of C_1 is CV_0
(C) the charge on the upper plate of C_2 is 0. (D) the charge on the upper plate of C_2 is $-CV_0$

CP0117

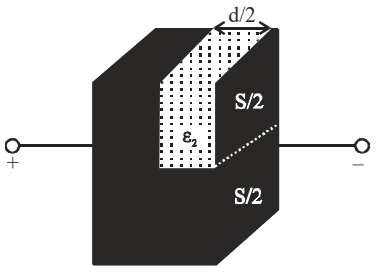
2. A parallel plate capacitor has a dielectric slab of dielectric constant K between its plates that covers 1/3 of the area of its plates, as shown in the figure. The total capacitance of the capacitor is C while that of the portion with dielectric in between is C_1. When the capacitor is charged, the plate area covered by the dielectric gets charge Q_1 and the rest of the area gets charge Q_2. The electric field in the dielectric is E_1 and that in the other portion is E_2. Choose the correct option/options, ignoring edge effects. **[JEE-Advance 2014]**



(A) $\frac{E_1}{E_2}=1$ (B) $\frac{E_1}{E_2}=\frac{1}{K}$ (C) $\frac{Q_1}{Q_2}=\frac{3}{K}$ (D) $\frac{C}{C_1}=\frac{2+K}{K}$

CP0118

3. A parallel plate capacitor having plates of area S and plate separation d, has capacitance C_1 in air. When two dielectrics of different relative permittivities ($\varepsilon_1 = 2$ and $\varepsilon_2 = 4$) are introduced between the two plates as shown in the figure,



the capacitance becomes C_2. The ratio $\frac{C_2}{C_1}$ is :-

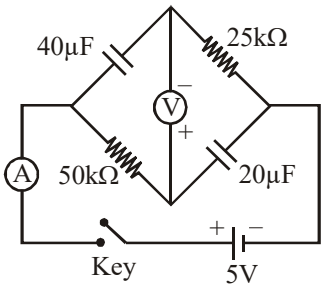
[JEE-Advance 2015]

(A) $\frac{6}{5}$ (B) $\frac{5}{3}$ (C) $\frac{7}{5}$ (D) $\frac{7}{3}$

CP0119

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\2-Electro-Grav-Current Elec.-Caps\Eng\04_Capacitance\01_[Eng].p65

4. In the circuit shown below, the key is pressed at time t = 0. Which of the following statement(s) is(are) true? **[JEE-Advance 2016]**



(A) The voltmeter displays –5V as soon as the key is pressed, and displays +5V after a long time
(B) The voltmeter will display 0 V at time t = ln 2 seconds
(C) The current in the ammeter becomes 1/e of the initial value after 1 second
(D) The current in the ammeter becomes zero after a long time

CP0120

PARAGRAPH–1

Consider a simple RC circuit as shown in figure 1.

Process 1 : In the circuit the switch S is closed at t = 0 and the capacitor is fully charged to voltage V_0 (i.e., charging continues for time T >> RC). In the process some dissipation (E_D) occurs across the resistance R. The amount of energy finally stored in the fully charged capacitor is E_C.

Process 2 : In a different process the voltage is first set to $\frac{v_0}{3}$ and maintained for a charging time T >> RC. Then the voltage is raised to $\frac{2v_0}{3}$ without discharging the capacitor and again maintained for a time T >> RC. The process is repeated one more time by raising the voltage to V_0 and the capacitor is charged to the same final voltage V_0 as in Process 1.

These two processes are depicted in Figure 2. **[JEE-Advance 2017]**

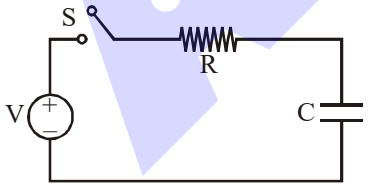


Figure 1

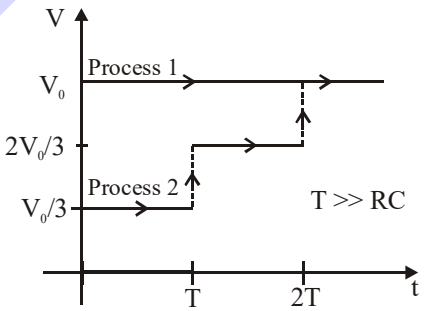


Figure 2

5. In Process 1, the energy stored in the capacitor E_C and heat dissipated across resistance E_D are related by :-

(A) $E_C = E_D$ (B) $E_C = 2E_D$ (C) $E_C = \frac{1}{2}E_D$ (D) $E_C = E_D \ln 2$

CP0121

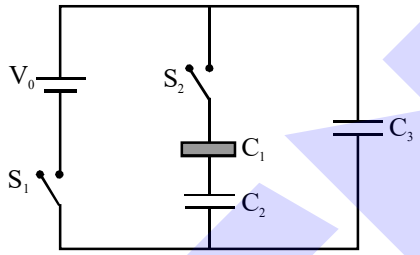
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Gravi-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

6. In Process 2, total energy dissipated across the resistance E_D is :-

(A) $E_D = \frac{1}{3}\left(\frac{1}{2}CV_0^2\right)$ (B) $E_D = 3\left(\frac{1}{2}CV_0^2\right)$ (C) $E_D = \frac{1}{2}CV_0^2$ (D) $E_D = 3CV_0^2$

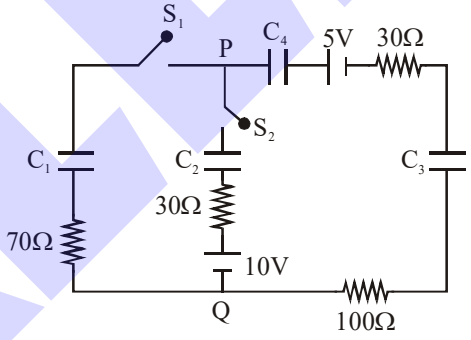
CP0122

7. Three identical capacitors C_1, C_2 and C_3 have a capacitance of 1.0 μF each and they are uncharged initially. They are connected in a circuit as shown in the figure and C_1 is then filled completely with a dielectric material of relative permittivity $\in_r$. The cell electromotive force (emf) $V_0 = 8V$. First the switch S_1 is closed while the switch S_2 is kept open. When the capacitor C_3 is fully charged, S_1 is opened and S_2 is closed simultaneously. When all the capacitors reach equilibrium, the charge on C_3 is found to be 5μC. The value of $\in_r$. **[JEE-Advance 2018]**



CP0123

8. In the circuit shown, initially there is no charge on capacitors and keys S_1 and S_2 are open. The values of the capacitors are $C_1 = 10$ μF, $C_2 = 30$ μF and $C_3 = C_4 = 80$ μF. **[JEE-Advance 2019]**



Which of the statement(s) is/are correct ?

(1) The keys S_1 is kept closed for long time such that capacitors are fully charged. Now key S_2 is closed, at this time, the instantaneous current across 30 Ω resistor (between points P and Q) will be 0.2 A (round off to 1st decimal place).

(2) If key S_1 is kept closed for long time such that capacitors are fully charged, the voltage difference between points P and Q will be 10 V.

(3) At time t = 0, the key S_1 is closed, the instantaneous current in the closed circuit will be 25 mA.

(4) If key S_1 is kept closed for long time such that capacitors are fully charged, the voltage across the capacitors C_1 will be 4V.

CP0124

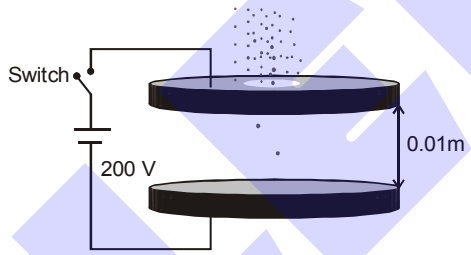
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy Module\2-Electro-Grav-Current Elec.-Caps\Eng\04_Capacitance\01_[Eng].p65

9. A parallel plate capacitor of capacitance C has spacing d between two plates having area A. The region between the plates is filled with N dielectric layers, parallel to its plates, each with thickness $\delta = \frac{d}{N}$. The dielectric constant of the mth layer is $K_m = K\left(1+\frac{m}{N}\right)$. For a very large N ($> 10^3$), the capacitance C is $\alpha\left(\frac{K \in_0 A}{d\ell n 2}\right)$. The value of α will be _____.
[$\in_0$ is the permittivity of free space] **[JEE-Advance 2019]**

CP0125

10. Two large circular discs separated by a distance of 0.01 m are connected to a battery via a switch as shown in the figure. Charged oil drops of density 900 kg m^{-3} are released through a tiny hole at the center of the top disc. Once some oil drops achieve terminal velocity, the switch is closed to apply a voltage of 200 V across the discs. As a result, an oil drop of radius 8×10^{-7} m stops moving vertically and floats between the discs. The number of electrons present in this oil drop is _____. (neglect the buoyancy force, take acceleration due to gravity = 10 ms^{-2} and charge on an electron (e) = 1.6×10^{-19} C)

[JEE-Advance 2020]

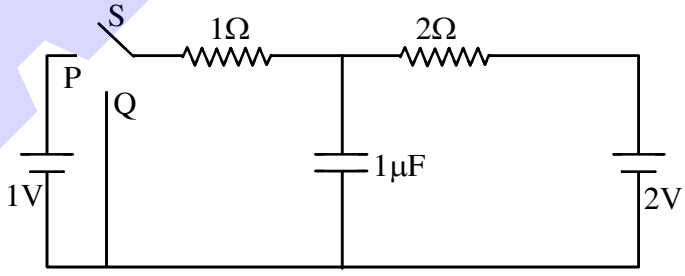


CP0145

Question Stem for Question Nos. 11 and 12

Question Stem

In the circuit shown below, the switch S is connected to position P for a long time so that the charge on the capacitor becomes q_1 μC. Then S is switched to position Q. After a long time, the charge on the capacitor is q_2 μC. **[JEE-Advance 2021]**



11. The magnitude of q_1 is ___ .

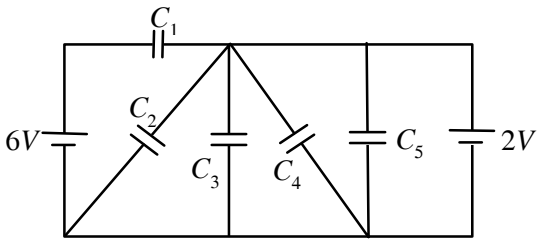
CP0146

12. The magnitude of q_2 is ___ .

CP0147

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Phy\Module\2-Electro-Gravi-Current Elec.-Capa\Eng\04_Capacitance\01_[Eng].p65

13. In the following circuit $C_1 = 12\ \mu F$, $C_2 = C_3 = 4\ \mu F$ and $C_4 = C_5 = 2\ \mu F$. The Charge stored in C_3 is _____ μC. **[JEE-Advance 2022]**



CP0207

14. A medium having dielectric constant K > 1 fills the space between the plates of a parallel plate capacitor. The plates have large area, and the distance between them is d. The capacitor is connected to a battery of voltage V. as shown in Figure (a). Now, both the plates are moved by a distance of $\frac{d}{2}$ from their original positions, as shown in Figure (b). **[JEE-Advance 2022]**

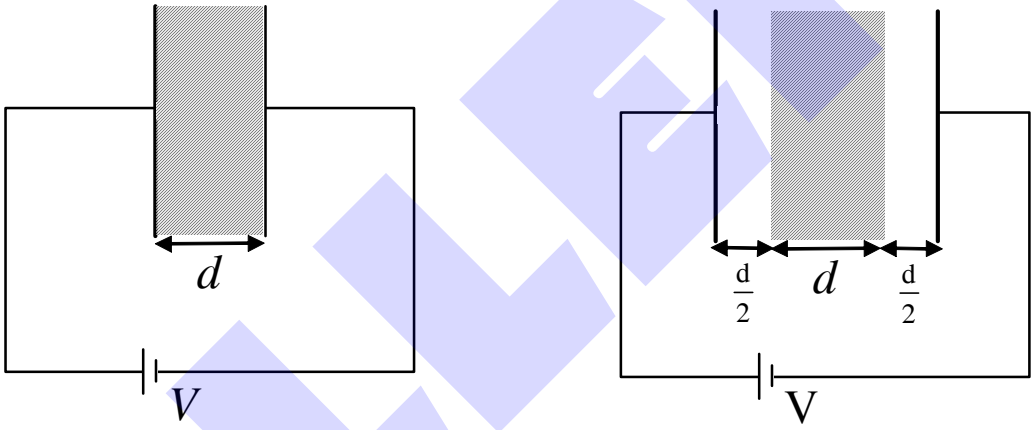


Figure (a) Figure (b)

In the process of going from the configuration depicted in Figure (a) to that in Figure (b), which of the following statement(s) is(are) correct?

(A) The electric field inside the dielectric material is reduced by a factor of 2K.

(B) The capacitance is decreased by a factor of $\frac{1}{K+1}$.

(C) The voltage between the capacitor plates is increased by a factor of (K + 1).

(D) The work done in the process **DOES NOT** depend on the presence of the dielectric material.

CP0208

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ANSWER KEY

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	B	D	C	C	D	A	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	B	D	D	A	A	D	A	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	D	D	B	C	B	C	D	B	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	B	A	C	B	D	A	B	B	C
Que.	41	42	43	44	45	46	47	48		
Ans.	C	D	C	B	D	D	D	A		

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	D	B	B	A	C	D	B	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	B	B	C	B	B	D	B,C,D	A,C,D
Que.	21	22	23	24	25	26	27	28		
Ans.	A,D	A,B,C	B,C	A,B,D	A,B	A,C	A,C	D		
Que.	29	A	B	C	D					
Ans.		P,Q,R	Q,S,T	S,T	P,R					

EXERCISE # S

1. 68 **2.** 5 **3.** 2.5 kV **4.** 9 **5.** 6V, 34V **6.** $\frac{32}{23}$ μF **7.** (C)

8. 30 V **9.** 12μC **10.** 3Q/2C **11.** $\frac{2A\varepsilon_0}{3d}$ **12.** 0

13. $\frac{25}{24}\frac{\varepsilon_0 A}{d}$ **14.** 4.425×10^{-9} Ampere **15.** $\frac{(K-1)CE}{2(K+1)}$ from C to B

16. $E\left[\frac{\varepsilon_0 ab(K-1)}{Md}\right]^{1/2}$ **17.** 100 V **18.** $\sqrt{6\rho L^5 \in_0 g}$ **19.** 225 **20.** 750

21. $Q_0 = \frac{CVR_2}{R_1+R_2}$ and $\alpha = \frac{R_1+R_2}{CR_1R_2}$ **22.** $\frac{3}{4}$J **23.** $\rho = \tau/\varepsilon_0\varepsilon \ln 2 = 1.4 \times 10^{13}$ Ω.m.

24. 2

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EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	4	4	3	4	1	2	4	Bonus
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	4	3	1	1	1	3	4	3	1
Que.	21	22	23	24	25	26	27	28		
Ans.	2	2	3	1	6	23	A	10		

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,D	A,D	D	A,B,C,D	A	A	1.50	3,4	1.00	6
Que.	11	12	13	14						
Ans.	1.33	0.67	8	B						

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Important Notes

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